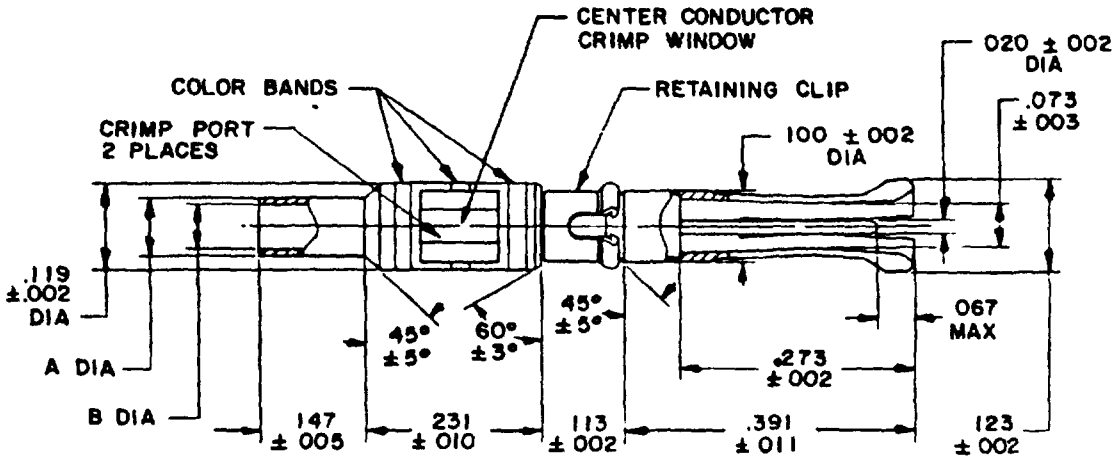
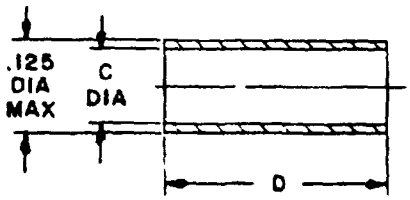
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, (FOR MIL-C-28748/10 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.





SHIELD CRIMP SLEEVE

Inches	mm	Inches	mm
.002	0.05	.100	2.54
.003	0.08	.113	2.87
.005	0.13	.119	3.02
.010	0.25	.123	3.12
.011	0.28	.125	3.18
.020	0.51	.147	3.73
.067	1.70	.231	5.87
.073	1.85	.273	6.93
		.391	9.93

TABLE I. Dimensions

BIN Code	A ± .002 (.05)	B ± .002 (.05)	C ± 0010 (025)	D ± .005 (.13)
438	.087 (2.21)	.070 (1.78)	1065 (2 705)	.305 (7.75)
439	.087 (2.21)	.045 (1.14)	.1035 (2 629)	250 (6.35)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Axial concentricity: All diameters to be concentric within .004 (.10 mm) TIR. After crimping, all diameters to be concentric within .009 (.23 mm) TIR.

FIGURE 1. Dimensions and configuration.

(A) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Dimensions, design characteristics and configuration. See figure 1 and table I.

Tools See table II.

Preparation of samples Contacts shall be wired as required using mating contact MIL-C-39029/79.

Wire in accordance with table I (cable accommodated column).

Operating temperature: 85°C maximum.

Tensile strength. See table III.

Contact engagement and separation forces See table III.

Contact resistance

Test current. 1.0 ampere.

Millivolt drop requirements: See table III.

Vibration. Method 2005 of MIL-STD-1344, test condition II.

Shock (specified pulse) Method 2004 of MIL-STD-1344, test condition I.

Dielectric withstanding voltage 750 V rms minimum at sea level.

Low signal level contact resistance (inner contact only).

Contact resistance (maximum)

Initial 100 milliohms.

After conditioning 125 milliohms.

Qualification by similarity. Qualification of one dash number (i.e. 16A) also qualifies all other dash numbers providing qualification samples are provided the qualifying activity.

Manufacturer's recommended assembly instructions to be shipped with unit package.

QPL evaluating activity Defense Electronics Supply Center, (DESC-E) Dayton, OH 45444.

Military part number. See table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.

TABLE I Design and characteristics.

BIN code	Color bands			Cables accommodated	Contact cavity size	Type	Class
	1st	2nd	3rd				
438	Yellow	Orange	Gray	M17/094-RG179 M17/113-RG316	16	D	A
439	Yellow	Orange	White	M17/093-RG178	16	D	A

TABLE II. Tools.

BIN code	Basic crimping tool	Installing tool	Removal tool
438 and 439	TMDC16CX4	M81969/17-04	TMC16RT008

Ⓐ TABLE III. Test limits.

Tensile strength (pounds)		Contact engaging and separation forces (oz)				Contact resistance (millivolts) 25°C + 3°C		Contact resistance (millivolts) 85°C + 3°C - 0°C		Cable used
Inner contact	Outer contact	Inner contact		Contact assembly 1/		Inner contact	Outer contact	Inner contact	Outer contact	
Min	Min	Min	Max	Min	Max	Max	Max	Max	Max	
4.0	10	0.4	7.0	1.5	20.0	8.0	4.0	12.0	4.0	RG316
4.0	10	0.4	7.0	1.5	20.0	12.0	4.0	12.0	4.0	RG178 RG179

1/ When mated with MIL-C-39029/79 contact.

TABLE IV. Military part number.

BIN code	Military part number	Supersedes part number
438	M39029/80-438	M39029/80-16A
439	M39029/80-439	M39029/80-16B

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AV
Navy - OS, SH, MC

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5935-3513-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:42Z
Check the source to verify that this is the current version before use.